

RUSSELLVILLE, AR, USA

WMO: 723429

Lat: 35.258N

Lon: 93.095W

Elev: 123

StdP: 99.86

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 53920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.7	(c) -5.8	(d) -15.2	(e) 1.0	(f) -3.1	(g) -12.8	(h) 1.3	(i) -2.4	(j) 9.1	(k) 7.3	(l) 8.2	(m) 5.9	(n) 1.2	(o) 320	(p) 0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.6	(c) 37.2	(d) 24.6	(e) 35.3	(f) 24.8	(g) 33.9	(h) 24.4	(i) 26.6	(j) 33.0	(k) 26.0	(l) 32.3	(m) 25.5	(n) 31.7	(o) 2.2	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.0	20.4	29.3	24.3	19.5	28.7	23.8	18.9	28.3	83.5	32.9	81.0	32.4	78.7	31.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.0	5.1	-12.0	38.5	2.5	2.7	-13.8	40.4	-15.3	42.0	-16.7	43.5	-18.5	45.5		
			-12.6	27.7	2.3	0.7	-14.3	28.2	-15.6	28.6	-16.9	29.0	-18.6	29.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	4.5	6.2	11.6	16.3	21.0	25.7	27.4	27.2	23.3	16.8	10.7	6.0		
	DBStd	9.24	5.38	5.42	5.19	4.30	3.76	2.56	2.48	2.81	3.49	4.65	4.94	4.92		
	HDD10.0	563	186	131	42	5	1	0	0	0	0	4	49	146		
	HDD18.3	1798	429	340	216	86	20	0	0	0	4	86	233	384		
	CDD10.0	2919	16	24	92	196	341	472	541	533	398	215	71	21		
	CDD18.3	1112	1	1	7	26	103	222	282	276	152	38	4	1		
	CDH23.3	11534	1	7	72	268	887	2302	3128	2969	1496	369	36	1		
(14) Wind	WSAvg	1.7	2.1	2.2	2.3	2.1	1.6	1.3	1.1	1.2	1.3	1.5	1.7	1.9		
	PrecAvg	1264	93	79	117	128	136	97	92	75	100	104	122	113		
	PrecMax	1731	186	206	353	295	322	214	168	177	230	273	309	258		
	PrecMin	913	18	22	41	44	14	32	31	16	16	7	26	6		
	PrecStd	224	49	44	69	61	72	40	35	39	51	64	82	56		
	DB	22.3	24.2	27.8	30.2	32.8	37.5	39.7	40.1	36.3	31.7	27.2	22.3			
	MCWB	16.4	16.8	18.0	20.5	22.4	23.3	23.7	24.7	23.3	21.6	19.7	17.7			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	19.8	21.2	25.6	28.2	31.1	35.0	37.0	37.3	34.0	29.6	23.9	19.7			
	2% MCWB	15.1	14.9	17.2	19.0	22.1	24.7	24.8	24.6	23.2	21.4	17.3	15.6			
	5% DB	17.2	18.7	23.0	26.5	29.7	33.5	35.1	35.3	32.4	27.4	21.8	17.2			
	5% MCWB	13.5	13.6	16.1	18.4	21.8	24.2	25.2	24.7	23.0	19.7	16.5	13.6			
	10% DB	14.0	16.2	20.9	24.5	28.0	32.1	33.6	33.7	30.6	25.1	19.5	14.4			
	10% MCWB	9.5	11.8	14.8	17.6	21.3	23.8	24.9	24.5	22.3	18.9	15.0	11.4			
	WB	18.3	18.4	19.8	22.3	24.7	26.8	27.5	27.3	25.5	24.0	20.9	18.7			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	20.7	22.0	25.5	27.8	29.8	33.0	33.3	34.1	31.8	29.2	24.7	21.3			
	2% WB	16.8	16.9	18.6	21.0	23.8	25.9	26.7	26.6	24.8	22.8	19.2	16.7			
	2% MCDB	19.2	19.9	23.9	25.6	28.6	32.2	32.8	33.3	30.9	27.2	22.5	19.2			
	5% WB	13.7	14.8	17.3	20.1	23.0	25.3	26.1	25.9	24.2	21.6	17.6	14.3			
	5% MCDB	16.0	17.9	21.6	24.5	27.7	31.2	32.2	32.5	29.6	25.3	20.3	16.3			
	10% WB	10.1	12.0	15.8	19.0	22.3	24.6	25.6	25.4	23.5	20.4	15.7	12.0			
	10% MCDB	13.4	15.3	19.7	23.1	26.7	30.1	31.5	31.7	28.5	23.7	19.0	13.9			
(35) Mean Daily Temperature Range	MDBR	11.8	11.7	13.0	12.9	11.5	11.5	11.6	11.7	12.9	13.1	12.8	11.1			
	5% DB MCDBR	16.0	16.2	15.5	15.2	13.0	13.1	13.6	14.1	14.4	15.0	15.2	14.4			
	5% DB MCWBR	10.8	10.3	8.6	7.5	5.6	4.6	3.9	4.2	5.0	7.1	9.5	10.3			
	5% WB MCDBR	11.8	12.4	12.8	11.7	10.8	11.3	11.4	11.9	11.9	11.8	12.5	11.6			
	5% WB MCWBR	9.7	9.4	8.0	6.8	5.4	4.6	4.3	4.3	4.7	6.2	9.2	9.6			
(40) Clear-Sky Solar Irradiance	taub	0.320	0.321	0.353	0.390	0.421	0.452	0.478	0.475	0.402	0.355	0.332	0.308			
	taud	2.493	2.489	2.428	2.329	2.303	2.231	2.204	2.188	2.378	2.487	2.506	2.542			
	Ebn at Noon	883	925	918	895	867	836	811	807	856	873	862	878			
	Edh at Noon	84	94	108	126	131	141	144	143	113	93	82	75			
(44) All-Sky Solar Radiation	RadAvg	2.48	2.97	4.08	5.12	5.59	6.34	6.19	5.72	4.90	3.74	2.67	2.16			
	RadStd	0.22	0.38	0.28	0.32	0.49	0.55	0.39	0.40	0.38	0.49	0.30	0.25			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+0.64	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page